

NAME: Diya S. Shah

## **Assignment: Windows Server – DNS, DHCP, File Sharing & Print Server**

### **1. Configure DNS Server and Create a Forward Lookup Zone**

Steps:

1. Open Server Manager.
2. Click Manage → Add Roles and Features.
3. Select DNS Server and install the role.
4. Open Tools → DNS.
5. Expand your server name.
6. Right-click Forward Lookup Zones → New Zone.
7. Select Primary Zone and click Next.
8. Enter the zone name as myapp.local.
9. Click Finish.
10. Right-click myapp.local → New Host (A or AAAA).
11. Enter:
  - Name: www
  - IP Address: 192.168.1.50
12. Click Add Host and then Done.

Result: The myapp.local forward lookup zone is created successfully, and the A record [www.myapp.local](http://www.myapp.local) points to 192.168.1.50.

### **2. Configure DHCP Server and Create a Scope**

Steps:

1. Open Server Manager.
2. Click Manage → Add Roles and Features.
3. Install the DHCP Server role.
4. Complete the DHCP Post-Install Configuration.
5. Open Tools → DHCP.
6. Expand your server name.

7. Right-click IPv4 → New Scope.
8. Enter the scope name as StudentScope.
9. Configure:
  - Start IP: 192.168.1.100
  - End IP: 192.168.1.120
10. Keep the default subnet mask (255.255.255.0).
11. Set the Lease Duration to 8 Hours.
12. Configure the default gateway if required.
13. Activate the scope.

Result: DHCP is configured successfully and assigns IP addresses from 192.168.1.100 to 192.168.1.120 with an 8-hour lease.

### **3. Share a Folder Named 'MusicLibrary'**

Steps:

1. Create a folder named MusicLibrary (for example, C:\MusicLibrary).
2. Right-click the folder → Properties.
3. Open the Sharing tab.
4. Click Advanced Sharing.
5. Check Share this folder.
6. Set the share name as MusicLibrary.
7. Click Permissions.
8. Remove Everyone.
9. Click Add and select the Students group.
10. Allow Read and Change permissions.
11. Click OK.
12. Open the Security tab.
13. Add the Students group.
14. Grant Read and Write permissions.
15. Click Apply and OK.

Result: Only users in the Students group can read and write files in the MusicLibrary shared folder.

## 4. Install Print Server Role and Share a Printer

Steps:

1. Open Server Manager.
2. Click Manage → Add Roles and Features.
3. Install the Print and Document Services role.
4. Select Print Server during installation.
5. After installation, open Devices and Printers.
6. Click Add Printer.
7. Select Microsoft XPS Document Writer.
8. Finish the installation.
9. Right-click the printer → Printer Properties.
10. Open the Sharing tab.
11. Check Share this printer.
12. Enter the share name CampusPrinter.
13. Click Apply and OK.

Result: The printer is shared as CampusPrinter and is available for network users.

## 5. Troubleshooting: Client PC Not Receiving an IP Address

### Cause 1: DHCP Server Service is Stopped

Test:

- Open **DHCP Console** or **Services (services.msc)**.
- Check whether the **DHCP Server** service is running.

Fix:

- Start the **DHCP Server** service.
- Restart the client PC and run:

```
ipconfig /release
```

```
ipconfig /renew
```

### Cause 2: DHCP Scope is Inactive or No Available IP Addresses

Test:

- Open **DHCP Console**.

- Verify that the scope is **Active**.
- Check whether free IP addresses are available.

**Fix:**

- Activate the scope if it is inactive.
- Expand the IP address range or remove unused leases if the pool is exhausted.

**Cause 3: Client Cannot Communicate with the DHCP Server**

**Test:**

- Check the network cable or VM network adapter.
- Run:

ping <DHCP\_Server\_IP>

- Verify both devices are on the same network.

**Fix:**

- Connect the client to the correct network.
- Ensure the network adapter is enabled.
- Restart the network adapter or reconnect the virtual network.